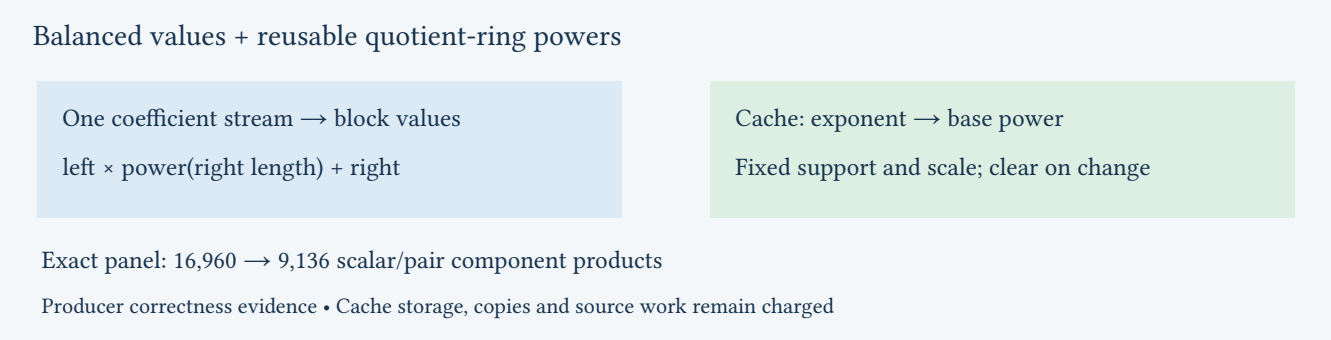
Reuse powers in split mode evaluation

T-1872 · 2026-11-07 · Prime-field compressed recovery

The evaluator now returns block values without recomputing a power at every node. Memoized binary exponentiation supplies the right-block power. One cache belongs to a fixed support/scale and can serve both C2 modes; a changed scale clears it.



Correctness. Induction on exponent proves cached powers; induction on block length proves the same streamed polynomial value. The predecessor remains a separate reference adapter.

Native control	Observed
Mode / independent power equalities	240 / 1,320
Reference / reused component products	16,960 / 9,136
Cold / warm products removed	2,416 / 5,408
Maximum mode cache entries / coefficient slots	13 / 134
Canonical state equalities	2,516
Source calls:281targets at cap64 / cap16	2,881 / 3,152

Four support shapes include unresolved-only, mixed, selected-only and empty;80scale resets are checked. Public recovery routes remain identical with zero unresolved targets. All inherited controls pass.

Costs. Value composition uses L−1 products plus cold power construction, conservatively O(log² L), per present component. Cache lookups, cloned values, capacities and map overhead remain charged. The134slots are live coefficient lengths, not peak allocated bytes. Product totals are structural kernel calls, not field-operation or timing ratios. Source grids, initial C0 and coefficient generation remain charged.

Failure retained. validation01 failed in the checker because an empty root list was passed to a nonempty product-tree API. validation02 uses the unit polynomial for empty support and passes; production evaluator unchanged. Both receipts and the failed checker snapshot are retained.

Evidence.72input pins per attempt; inspected PDF/SVG and publication manifest. Producer assertion; independent signed verdict pending. No cloud/timing run. Original4,563cohort unchanged. Full derivation/replay: DERIVATION.md, RESULTS.md, README.md.

Sources and next. Exact split predecessor; official FLINT quotient/powering conventions at flintlib.org/doc/nmod_poly/index.html. KB unavailable. Next share coefficient traversal across phase branches, accounting for simultaneous live state.